

JEN-WEI (Wayne) KUO 郭人瑋

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ABOUT ME

M.S. candidate in Computer and Communication Engineering at National Cheng Kung University, specializing in **AI/ML**, **computer vision**, **DRL-based optimization**, and **wireless communication systems**.

Kaggle Competitions Expert (1295/207932), Global Top 1%, with a **NeurIPS 2025 Silver Medal / Top 0.66% ranking** and **multiple AI competition Top-10% results**.

Experienced in PyTorch-based model development, feature engineering, computer vision, RAG/LLM applications, and edge AI deployment on Jetson/Raspberry Pi.

EDUCATION

National Cheng Kung University

09/2024 – 07/2026 (expected)

M.S. in Computer and Communication Engineering

- Advisor: Prof. Wenson Chang (張志文) — Intelligent Communication Lab
- **Research:** Voronoi-Based Energy-Aware User Partitioning and DRL Resource Allocation in Dual-UAV RIS-Assisted SWIPT Networks (time-switching · power allocation · RIS phase-shift) to energy-harvesting efficiency.

National Cheng Kung University

09/2020 – 06/2024

B.S. in Systems and Naval Mechatronic Engineering

- Capstone advisor: Prof. Chien-Hsing Lee (李建興), Lab617 — Intelligent Unmanned Vessel project (see below).

COMPETITIONS & AWARDS

- **Kaggle NeurIPS 2025 – Open Polymer Prediction Silver , Medal Top 0.66% (15 / 2,240)**
- **CVPR 2026 CV4CHL Workshop – The First AI for Children Challenge — Gold Award / 2nd Place (Team CCE)**
- **Kaggle Orbit Wars — Featured Simulation Competition , Bronze Medal Top 7.7% (365 / 4,729)**
- Kaggle Playground S5E8 – Binary Classification with a Bank Dataset — Top 6.7% (227 / 3,377)
- Kaggle Playground S6E3 – Predict Customer Churn — Top 6.9% (286 / 4,142)
- Taiwan Future Accident Classification Challenge (TAISC @ IEEE AVSS 2025) — Top 8.9% (12 / 134)
- AI CUP 2025 Spring – Table Tennis Smart Racket Analysis — Top 7.4% (47 / 633)
- 永豐 AI GO 股神對決 (Financial Forecasting) — Top 13.9% (121 / 868)

PUBLICATIONS

KUO Jen Wei, Yi-Ting Ku.

Clinically-Grounded Multi-View Features and Cross-Track Rule Coupling for Children Gait Visual Analysis.

Accepted as a non-proceeding submission, **CVPR 2026 Workshop on Computer Vision for Children (CV4CHL)**, 2026.

SELECTED PROJECTS

Intelligent Unmanned Vessel — Capstone Design Practicum

03/2023 – 01/2024

Role: Sound-source recognition · electromechanical integration | Stack: Raspberry Pi · Arduino Mega / UNO R4 Wi-Fi · Python · C++

- **Hull engineering:** Designed vessel in **Rhino + Orca 3D** (hydrostatics, FEA, resistance analysis) and built a balsa-wood hull with epoxy waterproofing and propulsion installation — drag values matched Holtrop theoretical estimates.
- **Autonomous heading control:** Implemented **PID + bang-bang closed-loop control** (MPU-9250 gyroscope + servo rudder) on separate MCUs to eliminate processing-delay conflicts.
- **Acoustic navigation:** Built a **real-time sound-source classification pipeline** (band-pass 500–3500 Hz → Log-MMSE denoising → cross-correlation matching) achieving accurate rising / flat / falling tone **recognition under motor noise**.
- **Computer vision:** Trained **YOLOv8** (3,000 frames, Tesla T4 GPU, precision 0.998 / AP@0.5 0.995) with **adaptive histogram equalization + Gaussian blur augmentation** for **underwater LED digit recognition**; resolved 2.5–3 s Pi↔Arduino USB latency by switching to **direct Pi servo control**.
- **Additional subsystems:** HC-SR04P ultrasonic obstacle avoidance and photoresistor light-source tracking; fully integrated into a sequential multi-modal pipeline.

AIDSP Edge AI & GenAI Deployment Suite

03/2026 – 06/2026

Stack: Jetson Xavier NX · FastAPI/Gradio · Docker · ONNX Runtime/TensorRT FP16 · YOLOv8-pose · Stable Diffusion/ControlNet · Gemma

- **Edge AI pipeline:** Built Jetson-targeted audio + LLM and real-time vision demos: 20 s audio capture -> log-mel -> ONNX genre inference -> Gemma lyrics, plus neural style transfer via ONNX/TensorRT FP16 with 15 FPS webcam streaming.
- **Gait + diffusion demos:** Built YOLOv8-pose gait authentication/tailgating alerts and a public ControlNet Canny/Sketch UI with scheduler, seed, CFG, weight controls, and ablation logging.

Brain Tumor MRI Segmentation — Attention U-Net + Elastic Augmentation

09/2025 – 11/2025

- Implemented custom Attention U-Net with elastic deformation augmentation for LGG MRI segmentation.
- **Achieved Val Dice 0.9070 / Dice 0.9116** — surpassing Kaggle U-Net baseline (Dice ≈ 0.896) with improved recall and boundary fit.

Unified-OneHead Multi-Task Learning Model

04/2025 – 06/2025

- Shared-encoder + single-head architecture covering classification, detection, and segmentation; **Replay Buffer** mitigates catastrophic forgetting — **performance degradation held within 5 % across all tasks.**

DynamicConv — Input-Conditioned Lightweight Convolution Module

03/2025 – 06/2025

- **MLP-generated dynamic kernels** supporting multi-channel input combinations (RGB / RG / R); outperforms static baselines in flexibility and inference efficiency.

TinyViT: Transformer-based Lightweight Vision Model

03/2025 – 05/2025

- **1.1 M parameters · 5 M FLOPs — matches ResNet-34 (90 % accuracy)**; ablation on FFN size and patch resolution validated design efficiency.

AI Mock-Interview Platform with GPT-4o & RAG

04/2025 – 06/2025

- **Streamlit + LangChain + OpenAI API**; **vector-DB-backed RAG** for résumé-aware Q&A; Dockerised for one-command deployment; **fine-tuned feedback** loop adapts to company culture profiles.

Traffic Accident Prediction — CLIP Semantic Features + XGBoost

04/2024 – 07/2024

- **Top 9.2 % (12/130) at IEEE AVSS TAISC 2025**; **CLIP-derived** semantic similarity features combined with **gradient-boosted trees** for accident-risk scoring from image data.

The controller design of the two-wheel balance car is based on the LQR control method

10/2022 – 01/2023

- **Arduino (C++) implementation**; MATLAB LQR simulation + **swarm-algorithm** parameter tuning for the inverted-pendulum stabilization task.

SKILLS

Programming: Python · C/C++ · MATLAB · FORTRAN

ML / CV: PyTorch · TensorFlow · scikit-learn · OpenCV · YOLOv8 · CLIP (VLM) · feature engineering

Deployment / MLOps: Docker · Streamlit · LangChain · API · RAG (vector DB) · Git · Linux ONNX Runtime · TensorRT

RL: DRL-based optimization · custom environment/simulator design

Wireless / DSP: Physical-layer communications · RIS · UAV communications · Signal processing

Embedded: Raspberry Pi · STM32 · Arduino (AVR / ARM) · NVIDIA Jetson Xavier / Xavier NX

Language: Mandarin (native) · English — TOEIC 720

RELEVANT COURSES

AI / ML	Deep Learning · Generative AI: Principles & Applications · Machine Learning · Computer Vision & Deep Learning · Generative AI Application System & Engineering · Artificial Intelligence in Digital Signal Analysis & Its Applications · Data Science · Digital Image Processing · Numerical Analysis
Communications / IC	Modern Communication · Wireless Networks & Applications · Communication Networks · Digital Signal Processing · Network Performance Analysis · Adaptive Signal Processing · Stochastic Processes · Digital / Analog Communications · Signals & Systems · Digital IC Design
CS	Computer Algorithms · Operating Systems · Computer Organization · Object-Oriented Programming · Parallel & Distributed Systems: Programming & Management
Control / Embedded	Automatic Control · Microcomputer Control & Embedded Systems

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Website: <https://kwannn666.github.io/>